ROOPA SREEDHAR A

PES1UG24CS386

SEC G

Github link: https://github.com/coderRoopa/PES1UG24CS386-pes-vcs

Screenshot 1A

```
roopa@LAPTOP-TVC42HGT:~/PES1UG24CS386-pes-vcs$ make test_objects
./test_objects                      # Screenshot 1A: all tests pass
./pes init
find .pes/objects -type f           # Screenshot 1B: sharded structure
gcc -Wall -Wextra -O2 -c test_objects.c -o test_objects.o
gcc -Wall -Wextra -O2 -c object.c -o object.o
gcc -o test_objects test_objects.o object.o -lcrypto
Stored blob with hash: d58213f5dbe0629b5c2fa28e5c7d4213ea09227ed0221bbe9db5e5c4b9aafc12
Object stored at: .pes/objects/d5/8213f5dbe0629b5c2fa28e5c7d4213ea09227ed0221bbe9db5e5c4b9aafc12
PASS: blob storage
PASS: deduplication
PASS: integrity check

All Phase 1 tests passed.
```

Screenshot 1B

```
.pes/objects/2a/594d39232787fba8eb7287418aec99c8fc2ecdaf5aaf2e650eda471e566fcf
.pes/objects/d5/8213f5dbe0629b5c2fa28e5c7d4213ea09227ed0221bbe9db5e5c4b9aafc12
.pes/objects/25/ef1fa07ea68a52f800dc80756ee6b7ae34b337afedb9b46a1af8e11ec4f476
roopa@LAPTOP-TVC42HGT:~/PES1UG24CS386-pes-vcs$
```

Screenshot 2A

```
roopa@LAPTOP-TVC42HGT:~/PES1UG24CS386-pes-vcs$ make test_tree
./test_tree                         # Screenshot 2A: all tests pass
gcc -Wall -Wextra -O2 -c index.c -o index.o
gcc -o test_tree test_tree.o object.o tree.o index.o -lcrypto
Serialized tree: 139 bytes
PASS: tree serialize/parse roundtrip
PASS: tree deterministic serialization

All Phase 2 tests passed.
```

Screenshot 2B:

```
roopa@LAPTOP-TVC42HGT:~/PES1UG24CS386-pes-vcs$ find .pes/objects -type f
.pes/objects/66/224663d23e6f4d9de9e2c7e6d8764305a92a3830a1a52d3d5f4aa8007b5c39
.pes/objects/e6/7ed66bcb4a708250a89f315d4d8bb92703343c84df834438880e13a68652bc
.pes/objects/14/da0e457e63669f91513f7c03f82dca4aa9bf28ffddc32501953ca05e399e67
.pes/objects/f1/0239b620372163c2dd06ac33189180c55edf1c56b15990a1cf44e00b1b0e59
.pes/objects/e0/0c50e16a2df38f8d6bf809e181ad0248da6e6719f35f9f7e65d6f606199f7f
.pes/objects/05/c8d777d0be85748b6788d1b5caf7d69754e6e68060c265ca21cbd6c05e5f43
.pes/objects/89/0e20721ba36faa9b0d56dc535f6201c349eb92f49686a3501fc7f03a94a7c6
.pes/objects/85/e367b76c2417b47a95f46208835ad8700dd58335171c94a497974be6191900
.pes/objects/33/b9aed2400f27a19f7d801d83267cd5b3da5b07fa1f9f852f7630b1877c005b
.pes/objects/2c/f8d83d9ee29543b34a87727421fdecb7e3f3a183d337639025de576db9ebb4
.pes/objects/96/c0d5c28dd3b719606c36723cd16aefbc73b3dc8b74de6107018335 7d6968b9
roopa@LAPTOP-TVC42HGT:~/PES1UG24CS386-pes-vcs$ xxd .pes/objects/66/224663d23e6f4d9de9e2c7e6d8764305a92a3830a1a52d3d5f4aa8007b5c39
00000000: 626c 6f62 2036 0048 656c 6c6f 0a         blob 6.Hello.
```

Screenshot 3A

```
roopa@LAPTOP-TVC42HGT:~/PES1UG24CS386-pes-vcs$ make pes
./pes init
echo "hello" > file1.txt && echo "world" > file2.txt
./pes add file1.txt file2.txt
./pes status          # Screenshot 3A
gcc -Wall -Wextra -O2 -c commit.c -o commit.o
gcc -Wall -Wextra -O2 -c pes.c -o pes.o
gcc -o pes object.o tree.o index.o commit.o pes.o -lcrypto
Initialized empty PES repository in .pes/
Staged changes:
  staged:     file1.txt
  staged:     file2.txt
```

Screenshot 3B

```
roopa@LAPTOP-TVC42HGT:~/PES1UG24CS386-pes-vcs$ cat .pes/index        # Screenshot 3B
100644 2cf8d83d9ee29543b34a87727421fdecb7e3f3a183d337639025de576db9ebb4 1776703118 6 file1.txt
100644 e00c50e16a2df38f8d6bf809e181ad0248da6e6719f35f9f7e65d6f606199f7f 1776703118 6 file2.txt
```

Screenshot 4A

```
roopa@LAPTOP-TVC42HGT:~/PES1UG24CS386-pes-vcs$ ./pes init
echo "Hello" > hello.txt && ./pes add hello.txt
./pes commit -m "Initial commit"
echo "World" >> hello.txt && ./pes add hello.txt
./pes commit -m "Add world"
echo "Goodbye" > bye.txt && ./pes add bye.txt
./pes commit -m "Add farewell"

./pes log                           # Screenshot 4A: 3 commits
Initialized empty PES repository in .pes/
Committed: 890e20721ba3... Initial commit
Committed: 14da0e457e63... Add world
Committed: 33b9aed2400f... Add farewell
commit 33b9aed2400f27a19f7d801d83267cd5b3da5b07fa1f9f852f7630b1877c005b
Author: Roopa Sreedhar A <PES1UG24CS386>
Date:   1776703237

    Add farewellauth!

commit 14da0e457e63669f91513f7c03f82dca4aa9bf28ffddc32501953ca05e399e67
Author: Roopa Sreedhar A <PES1UG24CS386>
Date:   1776703237

    Add worldellauth!

commit 890e20721ba36faa9b0d56dc535f6201c349eb92f49686a3501fc7f03a94a7c6
Author: Roopa Sreedhar A <PES1UG24CS386>
Date:   1776703237

    Initial committial commiA
```

Screenshot 4B:

```
roopa@LAPTOP-TVC42HGT:~/PES1UG24CS386-pes-vcs$ find .pes -type f | sort           # Screenshot 4B: object growth
.pes/HEAD
.pes/index
.pes/objects/05/c8d777d0be85748b6788d1b5caf7d69754e6e68060c265ca21cbd6c05e5f43
.pes/objects/14/da0e457e63669f91513f7c03f82dca4aa9bf28ffddc32501953ca05e399e67
.pes/objects/2c/f8d83d9ee29543b34a87727421fdecb7e3f3a183d337639025de576db9ebb4
.pes/objects/33/b9aed2400f27a19f7d801d83267cd5b3da5b07fa1f9f852f7630b1877c005b
.pes/objects/66/224663d23e6f4d9de9e2c7e6d8764305a92a3830a1a52d3d5f4aa8007b5c39
.pes/objects/85/e367b76c2417b47a95f46208835ad8700dd58335171c94a497974be6191900
.pes/objects/89/0e20721ba36faa9b0d56dc535f6201c349eb92f49686a3501fc7f03a94a7c6
.pes/objects/96/c0d5c28dd3b719606c36723cd16aefbc73b3dc8b74de61070183357d6968b9
.pes/objects/e0/0c50e16a2df38f8d6bf809e181ad0248da6e6719f35f9f7e65d6f606199f7f
.pes/objects/e6/7ed66bcb4a708250a89f315d4d8bb92703343c84df834438880e13a68652bc
.pes/objects/f1/0239b620372163c2dd06ac33189180c55edf1c56b15990a1cf44e00b1b0e59
.pes/refs/heads/main
```

Screenshot 4C:

```
roopa@LAPTOP-TVC42HGT:~/PES1UG24CS386-pes-vcs$ cat .pes/refs/heads/main          # Screenshot 4C part 1
33b9aed2400f27a19f7d801d83267cd5b3da5b07fa1f9f852f7630b1877c005b
roopa@LAPTOP-TVC42HGT:~/PES1UG24CS386-pes-vcs$ cat .pes/HEAD                     # Screenshot 4C part 2
ref: refs/heads/main
```

Final:

```
roopa@LAPTOP-TVC42HGT:~/PES1UG24CS386-pes-vcs$ make test-integration              # Final integration screenshot
=== Running integration tests ===
bash test_sequence.sh
=== PES-VCS Integration Test ===

--- Repository Initialization ---
Initialized empty PES repository in .pes/
PASS: .pes/objects exists
PASS: .pes/refs/heads exists
PASS: .pes/HEAD exists

--- Staging Files ---
Status after add:
Staged changes:
  staged:     file.txt
  staged:     hello.txt

Unstaged changes:
  (nothing to show)

Untracked files:
  (nothing to show)


--- First Commit ---
Committed: 0058097f0419... Initial commit

Log after first commit:
commit 0058097f04191af1cc35ce26141a2b49652911ecc82b62c3b1276ae9db15c463
Author: Roopa Sreedhar A <PES1UG24CS386>
Date:   1776703418

    Initial commit


--- Second Commit ---
Committed: 814e1a33ba54... Update file.txt

--- Third Commit ---
Committed: 237df4f992b6... Add farewell

--- Full History ---
commit 237df4f992b6fc18046e676bbf9fa8014e5a1c6b206df72469783308152c1405
Author: Roopa Sreedhar A <PES1UG24CS386>
Date:   1776703418

    Add farewellauth!

commit 814e1a33ba5465b0c4ba04f38091a5eb92044172b26e16d189207d61a0a27352
Author: Roopa Sreedhar A <PES1UG24CS386>
Date:   1776703418

    Update file.txth!

commit 0058097f04191af1cc35ce26141a2b49652911ecc82b62c3b1276ae9db15c463
Author: Roopa Sreedhar A <PES1UG24CS386>
Date:   1776703418

    Initial committial commiA


--- Reference Chain ---
HEAD:
ref: refs/heads/main
refs/heads/main:
237df4f992b6fc18046e676bbf9fa8014e5a1c6b206df72469783308152c1405

--- Object Store ---
Objects created:
10
.pes/objects/00/58097f04191af1cc35ce26141a2b49652911ecc82b62c3b1276ae9db15c463
.pes/objects/0b/d69098bd9b9cc5934a610ab65da429b525361147faa7b5b922919e9a23143d
.pes/objects/10/1f28a12274233d119fd3c8d7c7216054ddb5605f3bae21c6fb6ee3c4c7cbfa
.pes/objects/23/7df4f992b6fc18046e676bbf9fa8014e5a1c6b206df72469783308152c1405
.pes/objects/58/a67ed1c161a4e89a110968310fe31e39920ef68d4c7c7e0d6695797533f50d
.pes/objects/81/4e1a33ba5465b0c4ba04f38091a5eb92044172b26e16d189207d61a0a27352
.pes/objects/ab/5824a9ec1ef505b5480b3e37cd50d9c80be55012cabe0ca572dbf959788299
.pes/objects/b1/7b838c5951aa88c09635c5895ef7e08f7fa1974d901ce282f30e08de0ccd92
.pes/objects/d0/7733c25d2d137b7574be8c5542b562bf48bafeaa3829f61f75b8d10d5350f9
.pes/objects/db/07a1451ca9544dbf66d769b505377d765efae7adc6b97b75cc9d2b3b3da6ff

=== All integration tests completed ===
roopa@LAPTOP-TVC42HGT:~/PES1UG24CS386-pes-vcs$
```

Branching and Checkout

Q5.1: A branch in Git is just a file in .git/refs/heads/ containing a commit hash. Creating a branch is creating a file. Given this, how would you implement pes checkout <branch> — what files need to

change in .pes/, and what must happen to the working directory? What makes this operation complex?

To implement pes checkout <branch>:

- Read the target branch file (.pes/refs/heads/<branch>) to get its commit hash.
- Read that commit object and walk its tree to get the full file snapshot.
- Update every file in the working directory to match the tree contents (create, update, or delete).
- Update .pes/HEAD to "ref: refs/heads/<branch>\n".
- Rebuild the index to reflect the new HEAD's tree.

Complexity: if the user has modified tracked files, you must detect conflicts before touching anything . Also, files that exist in the current branch but not the target must be deleted — requiring careful bookkeeping.

Q5.2: When switching branches, the working directory must be updated to match the target branch's tree. If the user has uncommitted changes to a tracked file, and that file differs between branches, checkout must refuse. Describe how you would detect this "dirty working directory" conflict using only the index and the object store.

Walk every entry in the index. For each, compute its current on-disk blob hash (or use mtime+size as a cheap proxy). If any file's hash differs from what's in the index AND that file differs between the current and target branch trees, the checkout must refuse. The "conflict" is: index says file is at hash A, working dir has hash B, and target branch expects hash C where B ≠ A (meaning the user has local edits that would be overwritten).

Q5.3: "Detached HEAD" means HEAD contains a commit hash directly instead of a branch reference. What happens if you make commits in this state? How could a user recover those commits?

When HEAD contains a raw hash instead of a branch ref, new commits have no branch pointing to them — they dangle. To recover, the user can run pes branch rescue <hash> to create a new branch at that commit, provided they still have the hash. If they don't, they'd need to scan all objects (or the reflog in real Git) to find reachable but un-branched commits.

Garbage Collection and Space Reclamation

Q6.1: Over time, the object store accumulates unreachable objects — blobs, trees, or commits that no branch points to (directly or transitively). Describe an algorithm to find and delete these objects. What data structure would you use to track "reachable" hashes efficiently? For a repository with 100,000 commits and 50 branches, estimate how many objects you'd need to visit.

Use a mark-and-sweep approach: start from all branch refs and HEAD, walk every commit transitively (following parent pointers), then walk every tree in each commit (recursively), collecting every blob hash encountered. Store reachable hashes in a hash set. Then enumerate all objects under .pes/objects/ and delete any not in the set. For 100k commits and 50 branches: expect roughly 100k×(avg tree depth)×(avg files per dir) ≈ potentially millions of objects to visit; the hash set keeps membership tests O(1).

Q6.2: Why is it dangerous to run garbage collection concurrently with a commit operation? Describe a race condition where GC could delete an object that a concurrent commit is about to reference. How does Git's real GC avoid this?

Race: GC scans all reachable objects → decides blob X is unreachable → a concurrent pes add writes blob X to the store and an index entry pointing to it → GC deletes blob X → the next pes commit references a now-missing blob. Real Git avoids this by: (1) keeping a "grace period" — never deleting objects written in the last 2 weeks regardless of reachability; (2) using lock files; (3) considering any object written to the object store as temporarily "referenced" for the duration of GC.